

MTH211: Theory of Statistics.

End Semester Exam
2025-26 (II)
Solution:

1.

(a)

$$f(x) = \theta x^{\theta-1} \beta (1-x^{\theta})^{\beta-1}$$

$$= \theta x^{\theta-1} \beta \exp[(\beta-1) \ln(1-x^{\theta})]$$

$h(x) = \theta x^{\theta-1}$, $e(\beta) = \beta$, $\eta = \eta(\beta) = \beta-1$, $t(x) = \ln(1-x^{\theta})$, and
H: the natural parameter $= (-1, \infty)$ contains
an open rectangle of dim 1

$\therefore T(\underline{x}) = \sum_{i=1}^n \ln(1-x_i^{\theta})$ is a complete sufficient statistic.

(b)

$$L(\beta|\underline{x}) = \left(\theta^n \prod_{i=1}^n x_i^{\theta-1} \right) \beta^n \exp[(\beta-1) \sum_{i=1}^n \ln(1-x_i^{\theta})]$$

$$\ln L(\beta|\underline{x}) = \text{const.} + n \ln \beta + (\beta-1) \sum_{i=1}^n \ln(1-x_i^{\theta})$$

$$= \frac{\partial \ln L(\beta|\underline{x})}{\partial \beta} \bigg|_{\hat{\beta}} = \frac{n}{\hat{\beta}} + \sum_{i=1}^n \ln(1-x_i^{\theta}) = 0$$

$$\Rightarrow \hat{\beta} = \frac{-n}{\sum_{i=1}^n \ln(1-x_i^{\theta})}$$

$$\text{Further, } \frac{\partial^2 \ln L(\beta|\underline{x})}{\partial \beta^2} = -\frac{n}{\beta^2} < 0$$

$\Rightarrow \hat{\beta}$ is the unique MLE.

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$$2. f_X(x) = \frac{\theta x^{\theta-1}}{3^\theta}, \quad 0 < x < 3, \quad \theta > 0$$

$$(a) \quad E(X) = \frac{1}{3^\theta} \int_0^3 \theta x^\theta dx = \frac{3\theta}{\theta+1} = \mu, \text{ say.}$$

$$\Rightarrow E(\bar{x}) = \mu = \frac{3\theta}{\theta+1} \Rightarrow \frac{3\hat{\theta}_{M_0}}{\hat{\theta}_{M_0}+1} = \bar{x} = \frac{1}{n} \sum x_i \quad (\text{ie, } \hat{\mu}' = \mu')$$

$$\Rightarrow \hat{\theta}_{M_0} = \frac{\bar{x}}{3-\bar{x}}$$

$$(b) \quad \text{By WLLN, } \bar{x} \xrightarrow{P} \frac{3\theta}{\theta+1}$$

$$\Rightarrow \hat{\theta}_{M_0} = \frac{\bar{x}}{3-\bar{x}} \xrightarrow{P} \theta \Rightarrow \hat{\theta}_{M_0} \text{ is consistent for } \theta$$

$$(c) \quad \text{By CLT, } \sqrt{n}(\bar{x} - \mu) \xrightarrow{L} N(0, \sigma_{\bar{x}}^2)$$

$$\sigma_{\bar{x}}^2 = E(X^2) - E^2(X) = \frac{9\theta}{(\theta+2)(\theta+1)^2}$$

$$\text{By the A-method } \sqrt{n}(\hat{\theta}_{M_0} - \theta) \xrightarrow{L} N(0, \frac{\theta(\theta+1)^2}{\theta+2})$$

$$\text{Let } g(y) = \frac{y}{3-y}$$

$$\text{then } g'(y) = \frac{3}{(3-y)^2}$$

$$\left[\frac{g'(y)}{(g'(y))^4} \right]$$

$$g(\mu) = \frac{\frac{3\theta}{\theta+1}}{3 - \frac{3\theta}{\theta+1}} = \theta$$

$$y = \bar{x} \rightarrow g(y) = \frac{\bar{x}}{3-\bar{x}}$$

$$\text{from } \mu \rightarrow g(\mu) = \frac{\mu}{3-\mu}$$

$$\hookrightarrow \frac{\frac{3\theta}{\theta+1}}{3 - \frac{3\theta}{\theta+1}} = \theta$$

$$\text{from } \sigma_{\bar{x}}^2 \rightarrow (g'(\mu))^2$$

$$(g'(\mu))^2 = \frac{3}{(3 - \frac{3\theta}{\theta+1})^2} = \frac{(\theta+1)^2}{3}$$

$$\therefore \text{Var } \hat{\theta}_{M_0} = \frac{(\theta+1)^4}{9} \times \frac{9\theta}{(\theta+2)(\theta+1)^2}$$

$$= \frac{\theta(\theta+1)^2}{\theta+2}$$

$$2+3+5$$

$$\underline{6 \& 8}$$

(10)

$$f_X(x) = \frac{\alpha}{\theta^\alpha} x^{\alpha-1}, \quad 0 \leq x \leq \theta, \quad \theta, \alpha > 0.$$

$$L(\theta, \alpha | x) = \alpha^n \theta^{-n\alpha} \prod_{i=1}^n x_i^{\alpha-1} I_{(0, x_{(n)})} I_{(x_{(n)}, \theta)}.$$

For fixed α , $L(\theta, \alpha | x)$ is maximized for $\theta = x_{(n)}$
 $\Rightarrow \hat{\theta}_{MLE} = x_{(n)}$

$$\text{Then } \ln L(\alpha, \hat{\theta}_{MLE}) = n \ln \alpha - n\alpha \ln(\hat{\theta}_{MLE}) + (\alpha-1) \sum \ln(x_i).$$

$$\left. \frac{\partial}{\partial \alpha} \ln L(\alpha, \hat{\theta}_{MLE}) \right|_{\hat{\alpha}} = \frac{n}{\hat{\alpha}} - n \ln(\hat{\theta}_{MLE}) + \sum \ln(x_i) = 0$$

$$\Rightarrow \hat{\alpha} = \frac{n}{n \ln(\hat{\theta}_{MLE}) - \sum \ln x_i}$$

$$\text{Further, } \frac{\partial^2}{\partial \alpha^2} \ln L(\alpha, \hat{\theta}_{MLE}) = -\frac{n}{\alpha^2} < 0$$

$$\Rightarrow \hat{\alpha} \text{ is the unique MLE, } \hat{\alpha}_{MLE} = \frac{n}{n \ln(\hat{\theta}_{MLE}) - \sum \ln x_i}$$

$$(b) \quad F_X(x) = \int_0^x f_X(t) dt = \frac{\alpha}{\theta^\alpha} \int_0^x t^{\alpha-1} dt = \frac{\alpha}{\theta^\alpha} \cdot \frac{t^\alpha}{\alpha} = \left(\frac{x}{\theta}\right)^\alpha, \quad 0 \leq x \leq \theta.$$

$$\Rightarrow F_{X_{(n)}}(x) = [F_X(x)]^n = \left(\frac{x}{\theta}\right)^{n\alpha}$$

To satisfy, $P[aX_{(n)} \leq \theta \leq bX_{(n)}] = 0.90$, we calculate

$$P(aX_{(n)} \geq \theta) = 0.05 \quad \text{and} \quad P(bX_{(n)} \leq \theta) = 0.05$$

$$\Rightarrow P(X_{(n)} \geq \theta/a) = 0.05$$

$$\Rightarrow P(X_{(n)} \leq \theta/a) = 0.95$$

$$\Rightarrow a = \left(\frac{20}{19}\right)^{1/n\alpha}$$

$$\text{Similarly, } b = (20)^{1/n\alpha}$$

(10)

4. $f_X(x) = \frac{2x}{\theta} e^{-\frac{x^2}{\theta}}, x > 0, \theta > 0.$

(a) $\log f(x) = \log 2x - \log \theta - \frac{x^2}{\theta}$

$\frac{\partial}{\partial \theta} \log f(x|\theta) = -\frac{1}{\theta} + \frac{x^2}{\theta^2}$

$\frac{\partial^2}{\partial \theta^2} \log f(x|\theta) = \frac{1}{\theta^2} - \frac{2x^2}{\theta^3}$

$I_1(\theta) = -E\left[\frac{1}{\theta^2} - \frac{2x^2}{\theta^3}\right] = -\frac{1}{\theta^2} + \frac{2}{\theta^3} E(x^2) = -\frac{1}{\theta^2} + \frac{2}{\theta^2} = \frac{1}{\theta^2}$

=

[Let, $Y = X^2$, then $f_Y(y) = f_X(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} = \frac{1}{\theta} e^{-y/\theta} \Rightarrow Y \sim \text{Exp}(\theta)$. ^{Some may do this in part (b)}
 $\Rightarrow E(Y) = E(X^2) = \theta$]

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$\therefore \text{CRLB} = \frac{1}{n I_1(\theta)} = \frac{\theta^2}{n}$

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(b) 1-pr. exp. family

$f(x) = h(x) d(\theta) \exp(\eta T(x))$

with $h(x) = 2x, d(\theta) = \frac{1}{\theta}, \eta = \eta(\theta) = -\frac{1}{\theta}, T(x) = x^2$

$\Theta = \mathbb{R}^+$

Here the natural parameter space contains an open rectangle of dim. 1.

$\therefore T(x) = \sum X_i^2$ is a complete suff. for θ .

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And $E \sum X_i^2 = \sum E(X_i^2) = n\theta$

$\Rightarrow E\left(\frac{1}{n} \sum X_i^2\right) = \theta$

$\Rightarrow \frac{1}{n} \sum X_i^2$ is the C.S. stat. & unb. for θ

$\Rightarrow \frac{1}{n} \sum X_i^2$ is the UMVUE.

6+4

(10)

$$f(z|\theta) = \frac{1}{\theta^n} I(0, x_n) I(x_n, \theta)$$

$$\pi(\theta) = \beta \alpha^\beta \theta^{-(\beta+1)} I(\alpha, \theta)$$

$$\therefore f(x; \theta) \propto \beta \alpha^\beta \theta^{-(n+\beta+1)} I(\max\{x_n, \alpha\}, \alpha)$$

$$\Rightarrow \pi(\theta|x) \propto f(x; \theta)$$

with normalizing constant

$$f_x(x) = \int f(x; \theta) d\theta = \frac{\beta \alpha^\beta \{\max\{x_n, \alpha\}\}^{-(n+\beta)}}{n+\beta}$$

$$\text{and } \pi(\theta|x) = (\beta+n) \theta^{-(n+\beta+1)} [\max\{x_n, \alpha\}]^{\beta+n} I(\max\{x_n, \alpha\}, \infty)$$

which is same as $\pi(\theta)$ with
 $\beta^* = n+\beta$ and $\alpha^* = \max\{x_n, \alpha\}$

$$\therefore \text{posterior mean } E(\theta|x) = \frac{(\beta+n) \max\{x_n, \alpha\}}{\beta+n+1}$$

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$$f(z|\theta) = \frac{1}{\theta^n} I(0, x_n) I(x_n, \theta)$$

$$\pi(\theta) = \frac{1}{\sigma \sqrt{2\pi}} \exp\left\{-\frac{1}{2\sigma^2} (\ln \theta - \mu)^2\right\}$$

$$\pi(\theta|x) \propto \pi(\theta) f(z|\theta) = c \cdot \frac{1}{\theta^{n+1} \sigma \sqrt{2\pi}} \exp\left\{-\frac{1}{2\sigma^2} (\ln \theta - \mu)^2\right\} I(x_n, \theta)$$

$$\frac{\partial \ln \pi(\theta|x)}{\partial \theta} \bigg|_{\hat{\theta}_m} = -\frac{n+1}{\hat{\theta}_m} - \frac{(\ln \hat{\theta}_m - \mu)}{\hat{\theta}_m^2} = 0 \Rightarrow \hat{\theta}_m = \begin{cases} \exp(\mu - (n+1)\sigma^2) & \text{if } \exp(\mu - (n+1)\sigma^2) \geq x_n \\ x_n & \text{or} \end{cases}$$

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$$\therefore \text{posterior mode} = \max\left\{x_n, \exp\{\mu - (n+1)\sigma^2\}\right\}$$

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$$f(x|p) = p^{\sum x_i} (1-p)^{n-\sum x_i}$$

$$f(z, p) = f(z|p) \pi(p) = p^{\sum x_i} (1-p)^{n-\sum x_i} \cdot \frac{1}{B(\alpha, \beta)} p^{\alpha-1} (1-p)^{\beta-1}$$

$$= \frac{1}{B(\alpha, \beta)} \cdot p^{\alpha+\sum x_i-1} (1-p)^{n-\sum x_i+\beta-1}$$

$$\therefore f_X(z) = \frac{1}{B(\alpha, \beta)} \cdot B(\alpha+\sum x_i, n-\sum x_i+\beta)$$

$$\Rightarrow \pi(p|z) := B(\alpha+\sum x_i, n+\beta-\sum x_i)$$

$$\text{Hence, } E_{p|z} (p(1-p) | x_{2:n}) = \frac{1}{B(\alpha+\sum x_i, n+\beta-\sum x_i)} \int_0^1 p^{\sum x_i+\alpha} (1-p)^{n-\sum x_i+\beta} dp$$

$$= \frac{B(\alpha+\sum x_i+1, n-\sum x_i+\beta+1)}{B(\alpha+\sum x_i, n+\beta-\sum x_i)} = \frac{(\alpha+\sum x_i)(n+\beta-\sum x_i)}{(n+\alpha+\beta)(n+\alpha+\beta+1)}$$

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$$\delta_{\text{Bayes}} = \frac{(\alpha+1)(n+\beta-1)}{(n+\alpha+\beta)(n+\alpha+\beta+1)}, \quad T = \sum x_i$$

$$(b) f(z|p) = p^{\sum x_i} (1-p)^{n-\sum x_i}, \quad f(z, p) = p^{\sum x_i-1} (1-p)^{n-\sum x_i-1} \quad (\because \pi(p) = \frac{1}{p(p-1)})$$

$$\Rightarrow f_X(z) = B(\sum x_i, n-\sum x_i)$$

$$\therefore \pi(p|z) := B(\sum x_i, n-\sum x_i)$$

$$E_{p|z} [p(1-p) | x_{2:n}] = \frac{1}{B(\sum x_i, n-\sum x_i)} \int_0^1 p^{\sum x_i} (1-p)^{n-\sum x_i} dp = \frac{B(\sum x_i+1, n-\sum x_i+1)}{B(\sum x_i, n-\sum x_i)}$$

$$= \frac{\sum x_i (n-\sum x_i)}{n(n+1)} \quad \therefore \delta_{\text{Bayes}} = \frac{T(n-T)}{n(n+1)}, \quad T = \sum x_i$$

(Generalized) 8/

$$(c). f(z|p) = p^{\sum x_i} (1-p)^{n-\sum x_i}, \quad \pi(p) = 1$$

$$f(z, p) = p^{\sum x_i} (1-p)^{n-\sum x_i}, \quad \pi(p|z) = \frac{1}{B(\sum x_i+1, n-\sum x_i+1)} p^{\sum x_i} (1-p)^{n-\sum x_i}$$

$$\text{Weighted Bayes estimate} = \frac{E_{\pi(p)} [p(1-p) | x_{2:n}]}{E_{\pi(p)} [p | x_{2:n}]} = \frac{E \left[\frac{1}{1-p} | x_{2:n} \right]}{E \left[\frac{1}{p(1-p)} | x_{2:n} \right]}$$

$$E \left(\frac{1}{1-p} | x_{2:n} \right) = \frac{B(\sum x_i+1, n-\sum x_i)}{B(\sum x_i+1, n-\sum x_i+1)} = \frac{n+1}{n-\sum x_i}, \quad E \left[\frac{1}{p(1-p)} | x_{2:n} \right] = \frac{B(\sum x_i, n-\sum x_i)}{B(\sum x_i+1, n-\sum x_i+1)}$$

$$= \frac{(n+1)n}{\sum x_i (n-\sum x_i)}$$

$$\Rightarrow \text{the ratio} = \frac{\sum x_i}{n}$$

$$\delta_{NB} = \bar{X}$$

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The critical region is

$$w: \frac{f_1(x)}{f_0(x)} > k. \quad \Leftrightarrow 4x^{-3/2} > k.$$

$$\Leftrightarrow \alpha < c$$

$$\Leftrightarrow P_{H_0}(X < c) = \alpha \quad \text{ie,} \quad \frac{3}{64} \int_0^c x^2 dx = \alpha$$

$$\Rightarrow \frac{3}{64} \cdot \frac{c^3}{3} = \alpha$$

$$\Rightarrow c = 4\alpha^{1/3}$$

$$\begin{aligned} \text{Power of the test} &= P_{H_1}(X < c) = \frac{3}{16} \int_0^c x^{1/2} dx \\ &= \frac{3}{16} \cdot \frac{x^{3/2}}{3/2} \bigg|_0^c \\ &= \frac{1}{8} (4\alpha^{1/3})^{3/2} \\ &= \sqrt{\alpha} \end{aligned}$$

$$\therefore \alpha = 0.01 \quad \therefore \text{power} = 0.1$$

$$\underline{5+5} \quad (10)$$

8. $H_0: \delta \leq \delta_0$ ag. $H_A: \delta > \delta_0$

$$f(z; \delta, \theta) = \frac{1}{\theta^n} e^{-\frac{1}{\theta} \sum (x_i - \delta)} \cdot I(\delta, x_{(1)})$$

$$\lambda(z) = \frac{\sup_{\delta \leq \delta_0} L(\delta|z)}{\sup_{\delta \in R} L(\delta|z)} = \begin{cases} 1 & \text{if } x_{(1)} < \delta_0 \\ e^{-\frac{n}{\theta}(x_{(1)} - \delta_0)} & x_{(1)} \geq \delta_0 \end{cases}$$

$\therefore$ always accept for $x_{(1)} < \delta$ and reject if $\lambda(z) = e^{-\frac{n}{\theta}(x_{(1)} - \delta_0)} < c$
 $\Rightarrow x_{(1)} > d$, where d obtained from the size condition

$$P_{H_0}(x_{(1)} > d) = \alpha$$

$$\Rightarrow [P_{H_0}(x_i > d)]^n = \alpha$$

$$\text{and } P_{H_0}(x_i > d) = \alpha^{1/n}$$

$$\Rightarrow \int_d^{\infty} \frac{1}{\theta} e^{-\frac{(x - \delta_0)}{\theta}} dx = \alpha^{1/n} \Rightarrow d = \delta_0 - \frac{\theta}{n} \ln \alpha$$

Thus, the test ϕ is $\phi(z) = \begin{cases} 1 & \text{if } x_{(1)} > \delta_0 - \frac{\theta}{n} \ln \alpha \\ 0 & \leq \end{cases}$

9. $H_0: \delta = \delta_0$ ag. $H_A: \delta \neq \delta_0$

$$\lambda(z) = \frac{\sup_{\delta \neq \delta_0} L(\delta|z)}{\sup_{\delta \in R} L(\delta|z)} = \begin{cases} 0 & \text{if } x_{(1)} < \delta_0 \\ e^{-\frac{n}{\theta}(x_{(1)} - \delta_0)} & \geq \end{cases}$$

always reject if $x_{(1)} < \delta_0$ and conditionally when $e^{-\frac{n}{\theta}(x_{(1)} - \delta_0)} < c$, c by size condition.

$$\text{i.e., } x_{(1)} > d$$

where d is as obtained in part 8.

$$\therefore \phi(z) = \begin{cases} 1 & \text{if } x_{(1)} < \delta_0 \text{ or } x_{(1)} > \delta_0 - \frac{\theta}{n} \ln \alpha \\ 0 & \text{w} \end{cases}$$